

Christopher Francisque

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EDUCATION

University of Pennsylvania <i>MASc. Computer Science</i>	Philadelphia, PA June 2027
University of Massachusetts Amherst <i>B.S. Mechanical Engineering</i>	Amherst, MA June 2024

SKILLS

Languages: Python, C++, JavaScript/Typescript, MATLAB, Java, SQL, Bash
AI/ML: PyTorch, Transformers, Hugging Face, LLM Fine-tuning, LoRA, NumPy, Pandas
Tools: Google Cloud Platform, AWS, Git, Linux, Weights & Biases, Jupyter Notebook, Docker

EXPERIENCE

Pegasystems Waltham, MA
Product Manager May 2026 – Present

- Reduced incorrect AI responses by 20% by implementing a retrieval-augmented generation solution that grounds answers in official documentation rather than model priors
- Designing a self-improving agentic loop that iteratively evaluates its own outputs, incorporates feedback, and refines responses over time, increasing answer quality and reducing manual oversight

General Dynamics Mission Systems Dedham, MA
Software Engineer Jan 2026 – May 2026

- Built automated pipeline to parse C++ source and generate unit test plans leveraging the OpenAI API, eliminating ~80% of manual test-planning effort and saving 100+ engineering hours per month.
- Delivered a technical AI training to team on tools and workflows, contributing to a 2x increase in AI usage across the group.
- Built a Python-based key generation script to test device handling of encryption key formats, achieving 100% accuracy in 264 test vectors
- Created a GitLab CI/CD automation script that eliminated build failures on new runners

Brandeis University Waltham, MA
Machine Learning Research Assistant June 2025 – Jan 2026

- Implemented methods from 3 research papers, achieving 82-90% accuracy, while only training 0.037% of parameters
- Developed pruning pipelines using learned binary masks to identify redundant parameters, improving weak models by 4%
- Created a gradient tracking system to analyze parameter evolution during training, identified critical embedding layers whose removal caused complete model collapse despite being less than 0.01% of weights
- Built distributed training pipeline across 8 TPU cores handling 110M parameters and 67K samples, solved PyTorch XLA synchronization deadlocks that were causing training failure

Arcada San Francisco, CA
Software Engineer June 2025 – Aug 2025

- Identified security vulnerability by implementing Cloudflare infrastructure and rate limiting strategy, reducing bot attacks by 80%
- Scaled user growth to 100K+ unique users and generated 112K in revenue in 3 months from customers such as OpenAI and Google
- Owned end-to-end feature development for core platform capabilities including voting algorithm design and scroll pagination.
- Improved product performance by 30% through data-driven optimization initiatives; identified bottlenecks via user behavior analysis, defined requirements for database query improvements, and validated impact through A/B testing

Raytheon Andover, MA
Manufacturing Engineer June 2022 – June 2025

- Led many manufacturing improvement projects spanning root cause analysis, automation programming, fixture design, and training standardization, delivering \$160,000+ in cost savings, 10–30% cycle-time reductions, and a 25% cut in operator certification time
- Developed Python and C++ scripts to automate production data analysis for defect rates, operator performance, and statistical process control

PROJECTS

Self-Driving Toy Car Jan 2026 – Present

- Designed and integrated autonomous RC car using Raspberry Pi 5 and PCA9685 for precise PWM control of motors and servos
- Trained a convolutional neural network using PyTorch to map raw camera frames to continuous steering and throttle outputs, deploying the model for real-time inference on edge hardware
- Integrated camera capture, CNN inference, and servo actuation into a closed-loop control system at 15+ Hz with sub-220ms end-to-end latency